

## Question 2 - Hedge Fund Analysis

### Data Preparation

Before analysis, the data was cleaned by removing observations where factor values exceeded a magnitude threshold ( $|x| > 2$ ). These observations were treated as potential outliers and excluded to ensure the integrity of the analysis going forward.

### Q2.1. Multilinear Regression Model

A multilinear regression model was fitted using Ordinary Least Squares (OLS) to estimate the relationship between fund returns and risk factors. The dependent variable was the fund return series, while the independent variables were the provided factor returns.

The regression specification was:

$$R_t = \alpha + \beta_1 F_{1,t} + \beta_2 F_{2,t} + \dots + \beta_n F_{n,t} + \epsilon_t$$

where  $R_t$  represents hedge fund returns,  $\alpha$  represents fund alpha,  $\beta_i$  represents exposure to each factor, and  $\epsilon_t$  represents unexplained return.

### Significant Results

| Variable        | Beta    | p-value |
|-----------------|---------|---------|
| Alpha           | 0.0102  | 0.000   |
| Value vs Growth | -0.5729 | <0.001  |
| Credit          | 0.2419  | 0.011   |

### Model Selection

After fitting the full model, the statistical significance of factor exposures was assessed using their p-values at the 5% significance level.

A reduced model was therefore fitted using only these factors to avoid including variables that provide limited explanatory power.

| Model         | Adjusted $R^2$ | AIC    |
|---------------|----------------|--------|
| Full model    | 0.255          | -842.7 |
| Reduced model | 0.267          | -860.6 |

Although the full model achieved a higher  $R^2$ , the reduced model produced a higher **adjusted**  $R^2$  and a lower AIC. This suggests that, within this sample period, removing insignificant factors improves model efficiency by reducing unnecessary complexity.

The final model used in further analysis is therefore the reduced model:

$$R_t = 0.0093 - 0.5717F_{Value,t} + 0.1440F_{Credit,t} + \epsilon_t$$

where  $F_{Value,t}$  and  $F_{Credit,t}$  represent the Value vs Growth and Credit factor returns, respectively.

The positive alpha suggests the fund generated returns beyond what can be explained by its factor exposures, although this does not definitively prove manager skill as the estimate is sample dependent.

### Q2.2. Model Evaluation

| Metric         | Value |
|----------------|-------|
| $R^2$          | 0.275 |
| Adjusted $R^2$ | 0.267 |

| Metric           | Value  |
|------------------|--------|
| F-statistic      | 35.20  |
| F-test p-value   | <0.001 |
| Jarque–Bera test | 0.019  |

The model achieved an  $R^2$  of 0.275, meaning that approximately 27.5% of the variation in fund returns is explained by the selected factors. The remaining unexplained variation may represent manager skill, security selection, timing ability, or omitted risk factors.

The F-test was statistically significant ( $p < 0.001$ ), indicating that the model provides meaningful explanatory power.

The Jarque–Bera test ( $p = 0.019$ ) suggests mild departure from normality in the residuals, which is common in financial returns data.

Residuals were examined to evaluate model assumptions. The residuals appear centred around zero with no clear non-linear pattern, supporting the use of a linear regression model.

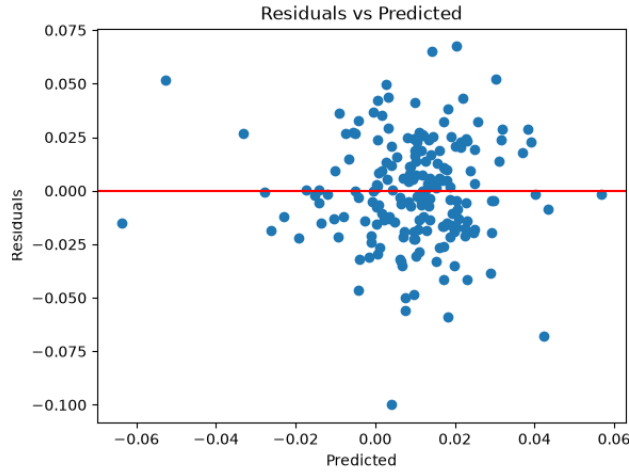


Figure 1: Residual Plot

The Breusch-Pagan test returned  $p = 0.953$ , providing no evidence of heteroskedasticity. Residual variance appears consistent across the sample period, supporting the use of standard errors.

The Q-Q plot shows residuals closely following the theoretical normal distribution across most of the range, with some deviation in the lower tail. This is consistent with the Jarque-Bera result and is typical of financial returns data where extreme negative events occur more frequently than a normal distribution would predict.

### Q2.3. Strategy Comparison

To investigate whether investing directly in the hedge fund was more profitable than replicating its factor exposure, a factor portfolio was constructed using the estimated betas from the reduced model.

$$R_{factor} = -0.5717F_{Value,t} + 0.1440F_{Credit,t} + \epsilon_t$$

Alpha was excluded because an independent investor cannot directly buy the manager's alpha.

The performance of both strategies was evaluated using the Sharpe ratio:

$$Sharpe = \frac{R}{\sigma}$$

where  $R$  represents the average return and  $\sigma$  represents return volatility. A risk-free rate of zero was assumed.

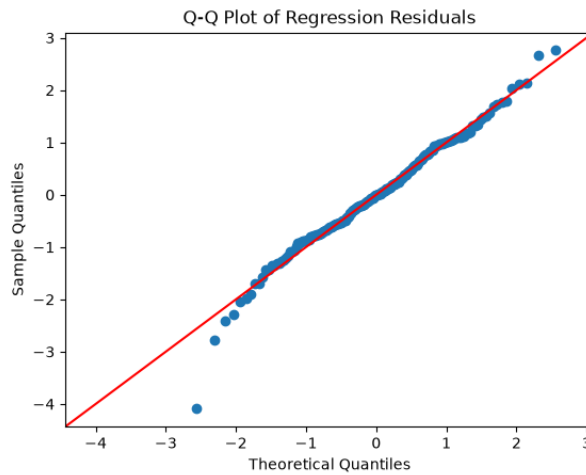


Figure 2: Q-Q Plot

| Strategy         | Average Return | Volatility | Sharpe Ratio |
|------------------|----------------|------------|--------------|
| Hedge Fund       | 0.0096         | 0.0288     | 0.335        |
| Factor Portfolio | 0.0003         | 0.0151     | 0.021        |

The hedge fund achieved a significantly higher Sharpe ratio than the factor portfolio. This indicates that the hedge fund generated higher risk-adjusted returns than simply investing in underlying factors.

This difference is consistent with the regression results, where the hedge fund displayed a positive and statistically significant alpha. The factor portfolio captures the systematic exposure to the Value vs Growth and Credit factors but does not capture the additional return represented by alpha.

Since alpha is positive and large enough, the fund remains superior before considering initial fees.

## Q2.4. Risk Comparison

Risk was evaluated using annualised volatility and maximum drawdown.

| Metric                | Hedge Fund | Factor Portfolio |
|-----------------------|------------|------------------|
| Annualised Volatility | 0.0997     | 0.0522           |
| Maximum Drawdown      | -0.1401    | -0.1731          |

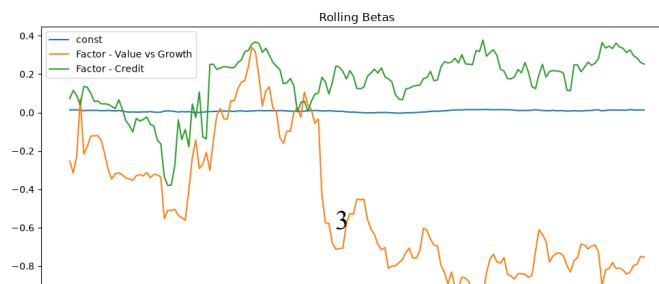
The hedge fund exhibited higher volatility, indicating greater variation in returns over time compared to the factor portfolio. However, the factor portfolio experienced a larger maximum drawdown, suggesting greater downside risk during periods of market stress.

This indicates that the two strategies differ in the type of risk they exhibit: the hedge fund is more volatile on a day-to-day basis, whereas the factor portfolio is more exposed to severe peak-to-trough losses.

Therefore, risk is not dominated by a single strategy and depends on whether volatility or downside risk is prioritised.

## Q2.5. Beta Stationarity

Rolling regressions were used to assess whether factor betas are stable over time.



| Factor          | ADF p-value | Conclusion     |
|-----------------|-------------|----------------|
| Value vs Growth | 0.528       | Non-stationary |
| Credit          | 0.076       | Non-stationary |

Both betas fail to reject the null hypothesis at the 5% level, confirming that factor exposures are non-stationary. Static betas from the reduced model therefore represent average exposure only and may misrepresent the fund's true risk profile at any given point in time.